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Optimizing performance of ultra-thin gas diffusion layer-less PEMFCs via tailored transition layer thickness between catalyst and metal foam

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Abstract

The drawbacks of conventional flow channel-rib flow fields and gas diffusion layers (GDLs) severely restrict mass transport and water management in proton exchange membrane fuel cells (PEMFCs), thereby limiting their volumetric power density. Our previous study proposed an ultrathin GDL-less PEMFC that uses metal foam to replace traditional flow fields and GDLs, significantly reducing mass transport distance and cell thickness while enhancing volumetric power density. To ensure contact and transition between the catalyst layer and metal foam, an ultrathin carbon nanofiber film (CNFF) is employed in this structure. This study systematically investigates the effect of CNFF thickness on the performance of ultrathin GDL-less PEMFCs. Results demonstrate that the protective effect of CNFF on the catalyst coated membrane (CCM) is strongly correlated with its thickness. Specifically, thinner CNFF offers less protection to the catalyst layer, resulting in an 30% difference in electrochemical active surface area (ECSA). A moderate increase in thickness reduces ohmic overpotential and enhances Knudsen diffusion within the oxygen catalyst layer, while excessive thickness leads to a decrease in oxygen molecular diffusion. Additionally, thicker CNFF provides better water storage and more effective water management under medium current densities, although performance degrades at ultrahigh current densities. Overall, the 25- μm CNFF balances these various factors to achieve the best integrated performance. These findings highlight that the optimal performance of GDL-less PEMFCs can be achieved by regulating the thickness of CNFF.

Keywords PEMFC, Metal foam, Flow field, Gas diffusion layer, Carbon nanofiber

1 Introduction

The increasing global energy demand and environmental challenges from fossil fuel use have made the transition to clean energy inevitable [1, 2]. Green hydrogen, produced from renewable resources, is particularly attractive for its high energy density and zero carbon emissions, and its integration with proton exchange membrane fuel cells (PEMFCs) enables efficient energy conversion and storage [3–6]. In PEMFCs, the bipolar plate is critical for electron and heat conduction, mechanical support, and reactant distribution, thus strongly affecting system cost, volume, and performance [7–9]. While conventional rib-channel flow fields have been optimized through various

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geometries and structural parameters, their inherent trade-offs between mass transport, pumping power, and manufacturability continue to limit further performance improvements [10–15].

In recent years, metal foam flow fields have emerged as a promising alternative to the conventional channel–rib configuration [16, 17]. Metal foam typically features high porosity, superior electrical and thermal conductivity, and great mechanical strength [18]. The porous skeleton provides sufficient mass transport path while facilitating rapid electron and heat conduction during cell operation. Moreover, the relatively mature manufacturing process, lower production costs, and lighter overall weight of metal foams make them attractive for use in PEMFCs and other electrochemical energy systems [19–21]. Previous studies have examined the intrinsic properties of metal foams, providing evidence of their feasibility and advantages in enhancing mass transfer and thermal management. Numerical simulations have shown that metal foams can enhance mass transport [22], improve reactant and product distribution [23], and achieve more uniform temperature profiles [24] under different design strategies. Experimental investigations further revealed that foam structural parameters [25] such as porosity, compression ratio, and surface modifications [26] and operating conditions such as back pressure, temperature, intake flow rate, and relative humidity strongly affect water–thermal management and durability. In addition, novel approaches such as porosity-gradient structures [27] and AI-based image analysis [28] have demonstrated the unique ability of metal foams to broaden the operational window, for example by enabling operation under a wider range of inlet humidity.

Utilizing the unique characteristics of metal foam, our research group proposed a novel concept of GDL-less PEMFC design [29]. In this approach, the conventional channel-rib flow field was replaced by a metal foam flow field, and an electrospun carbon nanofiber film (CNFF) was introduced as a multifunctional interlayer between the MEA and the flow field. This configuration eliminated the need for the traditional GDL. Our previous studies have demonstrated the significant potential of the GDL-less design in enhancing cell performance while reducing the overall volume. The volumetric power density was boosted by 111%. Furthermore, simulation results revealed that one of the key factors contributing to the significant performance enhancement of the GDL-less design is the substantial reduction in concentration losses, particularly under high current density conditions [30].

However, the thickness of CNFFs plays a critical role in determining their multifunctional properties. A thinner film can shorten the transport pathways, reduce

mass transport resistance, and thereby enhance volumetric power density. Nevertheless, excessive thinning may weaken mechanical integrity and lead to insufficient interfacial contact with the catalyst layer, which can cause local hot spots and durability issues. In contrast, a thicker CNFF improves structural robustness and interfacial conformity, but inevitably increases diffusion path length and hinders reactant accessibility. Moreover, CNFF thickness is closely associated with other key parameters, such as porosity distribution and compression behavior during cell assembly, creating a complex trade-off between mass transport, water management, and mechanical stability. Despite the critical influence of these coupled effects on PEMFC performance and durability, the impact of CNFF thickness has not yet been systematically investigated. Addressing this gap is essential for identifying the optimal thickness window for CNFFs, thereby advancing the rational design of GDL-less PEMFCs and guiding their practical implementation in high-power-density fuel cell systems.

Based on this understanding, the present study focuses on exploring the performance differences of GDL-less cells incorporating CNFFs of varying thicknesses. CNFFs with different thicknesses were fabricated and assembled into single cells. Their performance and internal mass transport characteristics were evaluated through polarization curves, electrochemical impedance spectroscopy (EIS), and microscopic ex-situ characterization. By combining experimental results with theoretical analysis, this study aims to elucidate the critical role of CNFF thickness in GDL-less configurations and provide valuable insights for enhancing the volumetric power density, reducing system complexity, and improving the overall manufacturability of PEMFCs.

2 Methods

2.1 Experiment preparation

2.1.1 CNFF preparation

To fabricate carbon nanofiber layers (CNFF) with uniform fiber diameter, porosity, and pore size distribution but varying thicknesses, key electrospinning parameters—spinning humidity, voltage, feed rate, and PAN solution concentration—are kept constant. By adjusting the total spinning time, CNFF samples with different thicknesses are successfully prepared. Table 1 lists the detailed electrospinning process parameters. Nanofiber membranes are fabricated with spinning times of 40 min, 80 min, 120 min, and 160 min. After curing at 240 °C and graphitization at 1000 °C, carbon nanofiber membranes with thicknesses of 8 µm, 15 µm, 25 µm, and 34 µm are obtained. Notably, the final CNFF thickness is not linearly proportional to the spinning time, as it is influenced by multiple electrospinning parameters, including

Table 1 Parameters for preparing CNFFs with different thicknesses by electrospinning

CNFF thickness	Spinning Duration (min)	Spinning Voltage (V)	Spinning Temperature (°C)	Ambient Relative Humidity (%)	Mass Fraction of PAN in Solution (%)
8 μm	40	12	25	40	14
15 μm	80	12	25	40	14
25 μm	120	12	25	40	14
34 μm	160	12	25	40	14

collector area, humidity, solution outlet rate, PAN concentration, and temperature. To minimize experimental errors, all CNFF samples in this study are continuously spun on the same day using PAN solutions from the same batch.

2.1.2 CCM Selection

The CCM (Catalyst Coated Membrane), the core component of PEMFC, consists of a proton exchange membrane and catalyst layers. As this study focuses primarily on the flow field and diffusion layer, the CCM parameters remain constant throughout the experiments. The selected proton exchange membrane is the M775.15 type from Gore (USA), a mechanically supported perfluorosulfonic acid membrane with a thickness of 15 μm. The cathode and anode catalyst layers employ XC-72 carbon support with platinum loadings of 0.4 mg·cm⁻² and 0.05 mg·cm⁻², respectively. The active area is 1 cm×3 cm. A 3 cm² activation area is chosen to mitigate the performance enhancement of the metal foam flow field in the planar direction, thereby highlighting the contribution of the GDL-free design in the thickness direction. During PEMFC assembly, the CCM and GDL/CNFF undergo no thermal pressing for lamination.

Notably, this experiment does not explore membrane or catalyst materials (i.e., the composite of the catalyst layer and proton exchange membrane). It merely demonstrates the advantages of the GDL-free structure under the condition of using the same CCM.

2.1.3 PEMFC assembly

The anode GDL uses Toray 060 carbon paper. The bipolar plate is constructed from graphite, with the flow field structure illustrated in Fig. S1. A PTFE film serves as the sealing gasket, employing surface sealing. The anode thickness is 150 μm, and the cathode thickness is 20 μm. The current collector plate consists of copper plated with gold, with a gold plating layer thickness of 0.3 μm. The end plate is made of aluminum alloy, and the bolt assembly preload is 5 N·m. These parameters are tabulated in Table S1. The structural schematics of different thicknesses are shown in Fig. 1.

2.2 Characterization methods

2.2.1 Polarization curve

The polarization curve measurements utilize the NBT100 test bench (Ningbo Beite Technology Co., Ltd.), featuring gas heating, humidification, backpressure control, and

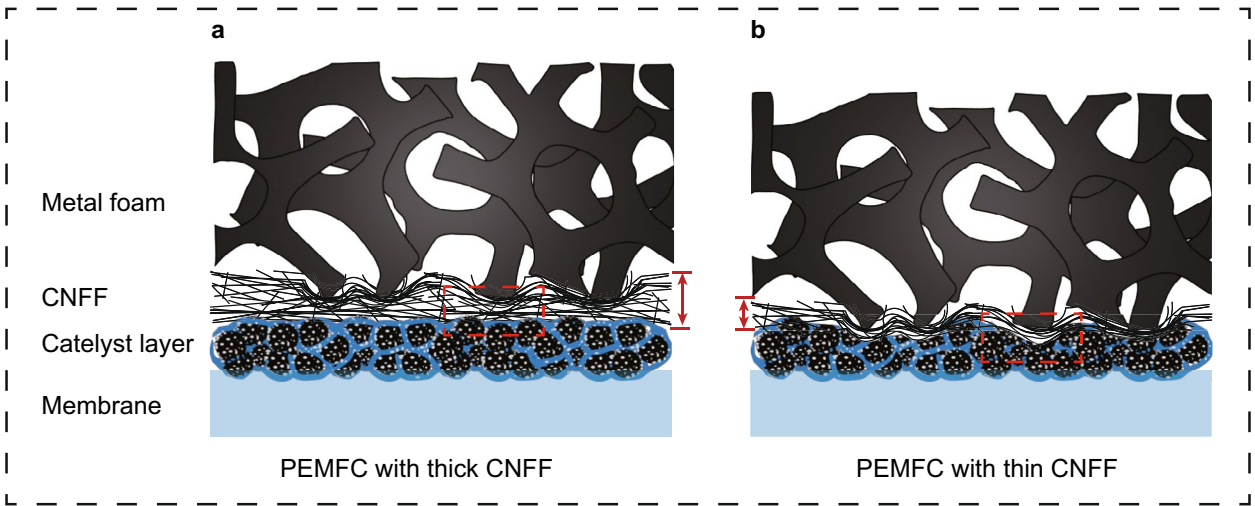


Fig. 1 The schematic diagrams of the internal structure of PEMFC when using (a) thick CNFF and (b) thin CNFF

end plate temperature regulation. Experimental conditions are as follows: Anode hydrogen (99.999% purity) and cathode air inlet temperatures are controlled at 70 °C, with flow rates of 0.2 SLPM and 0.5 SLPM, respectively. The PEMFC temperature is maintained at 70 °C via external heating wires. Inlet relative humidity for both electrodes is set to 100% using bubbling humidifiers. Measurements employ a constant-current mode with a data recording step of 0.05 A·cm⁻². Each working condition's polarization curve is measured in triplicate. All gases are supplied by Air Liquide (France).

2.2.2 Over potential decomposition

To elucidate the primary sources of polarization loss in PEMFCs with diverse structures, this study separates ohmic, activation, and concentration overpotentials using polarization curves and high-frequency impedance (HFR). The procedures are as follows:

The total polarization overpotential (η_{total}) is determined from the open-circuit voltage (E_{OCV}) and the actual output voltage (E_{cell}), as expressed in Eq. (1).

$$\eta_{\text{total}} = E_{\text{OCV}} - E_{\text{cell}} \quad (1)$$

where E_{OCV} and E_{cell} are in volts (V). The total polarization overpotential comprises the ohmic overpotential (η_{ohmic}), activation overpotential (η_{act}), and concentration overpotential (η_{con}), as shown in Eq. (2):

$$\eta_{\text{total}} = \eta_{\text{ohmic}} + \eta_{\text{act}} + \eta_{\text{con}} \quad (2)$$

The ohmic overpotential is calculated using Eq. (3):

$$\eta_{\text{ohmic}} = R_{\text{HFR}} \cdot i \quad (3)$$

where R_{HFR} (Ω) is the ohmic resistance measured by HFR at the corresponding current density, and i (A·cm⁻²) is the operating current density. The activation overpotential (η_{act}) is derived from the Tafel equation, presented in Eq. (4):

$$\eta_{\text{act}} = \alpha \ln i + b \quad (4)$$

where a is the Tafel slope and b is the intercept. At ultra-low current densities (50 mA·cm⁻² to 120 mA·cm⁻²) the concentration overpotential is negligible. Thus, the polarization overpotential excluding the ohmic component ($\eta_{\text{total}} - \eta_{\text{ohmic}}$) shows a linear relationship with $\ln i$. Consequently, η_{act} in Eq. (2) can be omitted, simplifying Eq. (2) to Eq. (5):

$$\eta_{\text{total}} \approx \eta_{\text{ohmic}} + \eta_{\text{act}} = R_{\text{HFR}} \cdot i + (\alpha \ln i + b) \quad (5)$$

The Tafel slope a and intercept b are determined by fitting η_{total} and η_{ohmic} data within the 50 mA·cm⁻²

to 120 mA·cm⁻² range. Subsequently, η_{act} across the full current density spectrum is calculated using Eq. (4), and η_{con} is obtained by rearranging Eq. (2). This process completes the separation of the three polarization overpotentials.

2.2.3 Electrochemical impedance spectroscopy

In this experiment, electrochemical impedance spectroscopy (EIS) of the PEMFC is measured using a Zahner Zennium Pro electrochemical workstation (Germany). The test protocol is as follows: The PEMFC is connected to the workstation and operated at a specified current density under galvanostatic conditions identical to those in polarization curve tests. After allowing sufficient time for voltage stabilization at the target current density, measurements are conducted in galvanostatic mode with an AC perturbation signal amplitude of 5%–10% of the DC current. The frequency range spans 100 mHz to 100 kHz. Specific measurement parameters are tabulated in Table S2.

2.2.4 Distribution of relaxation times

Distribution of Relaxation Times (DRT) analysis is employed to further interpret the EIS. This method transforms the frequency-domain impedance data into a distribution function of relaxation times, thereby distinguishing overlapping electrochemical processes such as charge transfer, mass transport, and ohmic contributions. Prior to the analysis, the data were validated by the Kramers–Kronig relations to ensure causality, linearity, and time invariance. The DRT approach allows the experimentally measured spectra to be deconvoluted without relying on a specific equivalent circuit model, offering a more detailed and physically meaningful characterization of polarization resistances. The impedance value R of the electrochemical process corresponding to each peak in DRT is calculated by integrating $\gamma(\tau)$ with respect to $\ln \tau$ [31]. A full description of the theoretical framework, mathematical formulation, and numerical implementation used in this study is provided in the Supplementary Information Statement S1.

The software utilized for DRT analysis of PEMFC electrochemical impedance spectra in this study is "DRT tools", publicly available from the Ciucci Research Group at the Hong Kong University of Science and Technology.

2.2.5 Oxygen transport resistance

In this experiment, the cathode is supplied with a mixed gas containing 1% dry molar fraction of oxygen (x_{O_2}) in air and nitrogen (99.999% purity, Air Liquide, France), while the anode is supplied with hydrogen (99.999% purity, Air Liquide, France). The test conditions are listed in Table S3. A polarization curve is

obtained, and the current density at 0.1 V is defined as the limiting current density (i_{lim}). The oxygen diffusion resistance is then calculated using this limiting current density. At the limiting current density, the oxygen diffusion resistance R_p (unit: $\text{s}\cdot\text{cm}^{-1}$) for a specific oxygen molar fraction x_{O_2} and cathode pressure p_{Ch} is calculated by Eq. (6):

$$R_t = \frac{4Fx_{\text{O}_2}(P_{\text{Ch}} - P_w)}{RT} \times \frac{1}{i_{\text{lim}}} \quad (6)$$

where p_{Ch} is the absolute pressure in the cathode flow channel (kPa), p_w is the water vapor partial pressure in the cathode supply mixer (kPa), R is the universal gas constant ($8.314 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$), F is the Faraday constant, T is the fuel cell temperature (K), x_{O_2} is the oxygen molar fraction in the cathode-supplied N_2/O_2 mixture, and i_{lim} is the limiting current density ($\text{mA}\cdot\text{cm}^{-2}$).

The oxygen transport resistance consists of a pressure-dependent term and a pressure-independent term. The pressure-dependent term R_p primarily originates from molecular diffusion in the porous medium, while the pressure-independent term R_{NP} mainly includes Knudsen diffusion in the catalyst layer micropores,

transport through the ionomer, and transport in liquid water. To separate these terms, the limiting current density of each sample is measured under four different chamber pressure conditions, and R_t is calculated. Finally, r_p and R_{NP} are obtained by fitting Eq. (7):

$$R_t = R_{\text{NP}} + R_p = R_{\text{NP}} + r_p \cdot p_{\text{Ch}} \quad (7)$$

In Eq. (12), R_{NP} is the pressure-independent term of oxygen transport resistance ($\text{s}\cdot\text{cm}^{-1}$), R_p is the pressure-dependent term ($\text{s}\cdot\text{cm}^{-1}$), and r_p is the pressure-dependent term coefficient ($\text{s}\cdot\text{cm}^{-1}\cdot\text{kPa}^{-1}$).

3 Result and discussion

3.1 Influence on the electrochemical characteristics

Four CNFF thicknesses (34 μm , 25 μm , 15 μm , and 8 μm) are tested for performance. Figure 2a and b show the polarization curves and IR-free curves of ultra-thin GDL-less PEMFCs with varying CNFF thicknesses. The maximum power density first increases and then decreases as CNFF thickness decreases from 34 μm to 8 μm . At 34 μm , the maximum power density is $1.08 \text{ W}\cdot\text{cm}^{-2}$; when the thickness is reduced to 25 μm , it increases to $1.31 \text{ W}\cdot\text{cm}^{-2}$ (a 21.3% improvement), likely due to shorter oxygen transport pathways and enhanced reactive gas

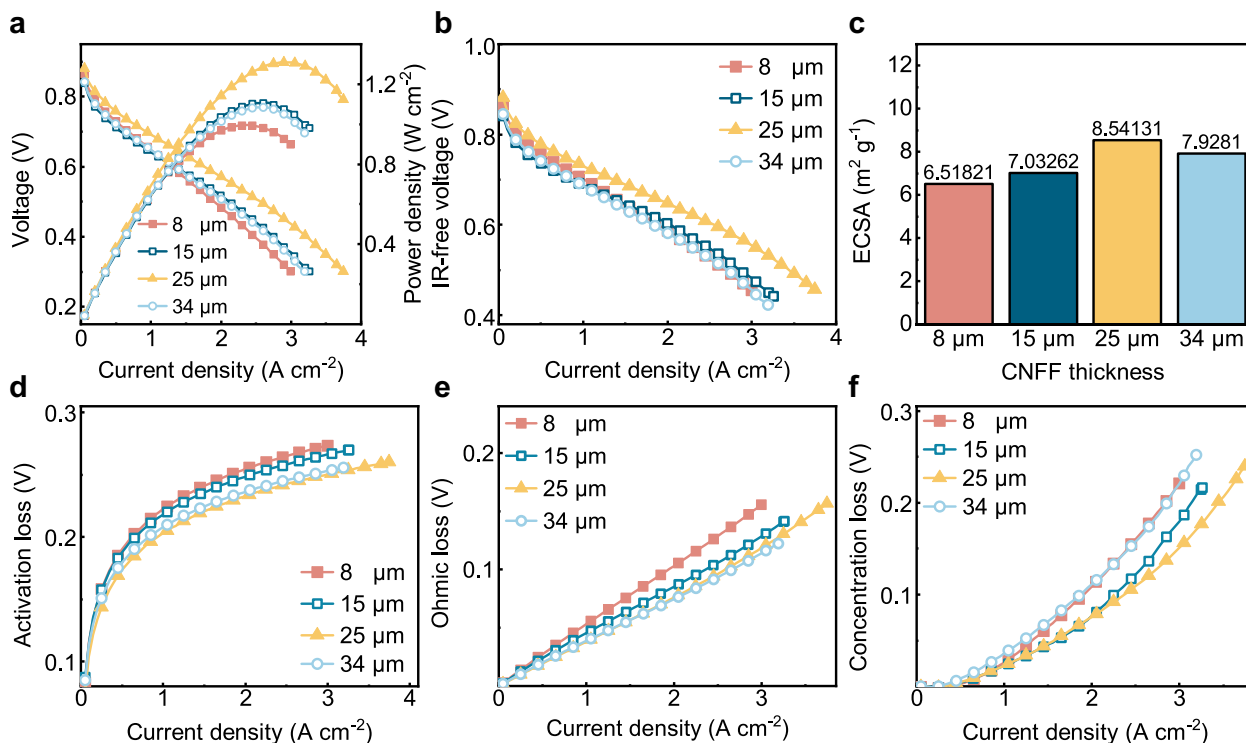


Fig. 2 The thickness of CNFF has a significant impact on the electrochemical performance of ultrathin GDL-less PEMFC. **a** Polarization curves without backpressure for operation conditions given in Table S4. **b** IR-Free voltage (polarization curves excluding the ohmic overpotential), **(c)** Electrochemical reaction area. The variations of **(d)** Activation overpotential, **(e)** ohmic overpotential and, **(f)** concentration overpotential with current density

diffusion efficiency resulting from the reduced thickness. Further thinning causes performance degradation: at 15 μm , the maximum power density is $1.11 \text{ W}\cdot\text{cm}^{-2}$, and at 8 μm , it drops to $1.0 \text{ W}\cdot\text{cm}^{-2}$, representing 15.3% and 23.7% decreases compared to 25 μm . This decline may arise because decreasing CNFF thickness shortens reaction gas transport distances but weakens its buffer effect on the metal foam skeleton, damaging the catalyst layer at skeleton positions, reducing the three-phase interface, and exacerbating activation losses. Additionally, overly thin CNFF enhances lateral electron transfer in the catalyst layer, worsening electronic ohmic losses. Severe CNFF compression at skeleton positions also causes mass transfer limitations, contributing to performance decline.

To clarify performance degradation mechanisms at reduced CNFF thicknesses, separated activation, ohmic, and concentration overpotentials are analyzed. Figure 2d shows the activation overpotentials for the four CNFF thicknesses. Below 25 μm , the activation overpotential decreases with increasing CNFF thickness. Thin CNFF provides inadequate catalyst layer protection, allowing nickel foam skeletons to damage the three-phase reaction interface and worsening activation overpotentials. ECSA tests (Fig. 2c) confirm that the PEMFC with 8- μm CNFF has the lowest ECSA, which increases as the thickness increases from 8 μm to 25 μm , thus validating the above conclusion. When the thickness increases from 25 μm to 34 μm , the activation overpotential slightly increases, indicating that 25 μm CNFF provides optimal protection to the catalyst layer. In addition, the performance comparison between the GDL-less PEMFC with a 34 μm -thick CNFF and the conventional parallel flow field PEMFC with GDL is presented in Fig. S2. The structural parameters of the conventional PEMFC are provided in Table S5.

As depicted in Fig. 2e, the ohmic overpotential does not decrease with the reduction in CNFF thickness. This phenomenon can be attributed to two key factors: first, thinner CNFF inherently exhibits inferior lateral electron conductivity; second, as illustrated in Fig. 1, thinner CNFF is more susceptible to compression-induced fracture by the metal foam, thereby resulting in increased contact resistance.

Figure 2f shows the concentration overpotentials, which first decrease then increase as the CNFF thickness gradually increases, primarily influenced by oxygen transport resistance. When the CNFF thickness increases from 8 μm to 15 μm , the concentration overpotential decreases significantly, likely for two main reasons. Firstly, as illustrated in the schematic of Fig. 1, thinner CNFF allows easier intrusion and damage to the catalyst layer by the metal foam skeleton, damaging nanopores or polymers within the catalyst layer and increasing the

difficulty of oxygen transport to the three-phase interface. Secondly, compression damage from the metal foam skeleton reduces the porosity of CNFF at skeleton-aligned positions; thinner CNFF has poorer resistance to such compression, leading to severe porosity reduction at skeleton-aligned areas and affecting the porosity at the CNFF-catalyst layer interface, ultimately decreasing oxygen transport capacity. The first scenario increases the pressure-independent term of oxygen transport resistance, while the second increases the pressure-dependent term. When the thickness of CNFF increases from 25 μm to 34 μm , the concentration overpotential rises again. The main reason is that when the thickness of CNFF reaches a certain level, its protective effect on the catalyst layer has already reached a relatively high degree (both the ECSA and the SEM images of the catalyst layer confirm this). Therefore, the impact of the increased oxygen transport distance caused by the greater thickness becomes prominent, leading to a rise in concentration loss.

To validate that the metal foam skeleton may damage the catalyst layer's three-phase interface when the CNFF is too thin, ex-situ characterizations of the catalyst layer were conducted after cell operation using CNFFs with thicknesses of 8 μm and 34 μm . The SEM results are shown in the Fig. 3. As can be clearly observed from Fig. 3a, the catalyst layer surface with the 8- μm CNFF exhibits severe damage caused by excessive contact with the foam skeleton, and the magnified image (Fig. 3c-f) further reveals large CNFF residues trapped in the dents. In contrast, the catalyst layer surface with the 34- μm CNFF shows far fewer and milder damages (Fig. 3b), indicating that increasing the CNFF thickness can effectively mitigate the skeleton-induced destruction of the three-phase interface.

3.2 Oxygen transport resistance

Figure 4a presents the oxygen transport resistance of PEMFCs with different thicknesses of CNFF (The corresponding limiting current density is shown in Fig. S3, and Table S6). Taking a chamber pressure of 150 kPa as an example, the total oxygen transport resistance decreases from $0.595 \text{ s}\cdot\text{m}^{-1}$ at 34 μm to $0.531 \text{ s}\cdot\text{m}^{-1}$ at 25 μm , attributed to the shortened oxygen diffusion distance enhancing diffusion efficiency according to Fick's law [32]. Further reducing the CNFF thickness to 15 μm increases the total oxygen transport resistance despite the shorter diffusion path, consistent with the concentration overpotential trend observed in the previous section. To elucidate this phenomenon, the total resistance is decomposed into pressure-dependent and pressure-independent components.

Figure 4b shows these components for different CNFF thicknesses. The pressure-dependent term increases

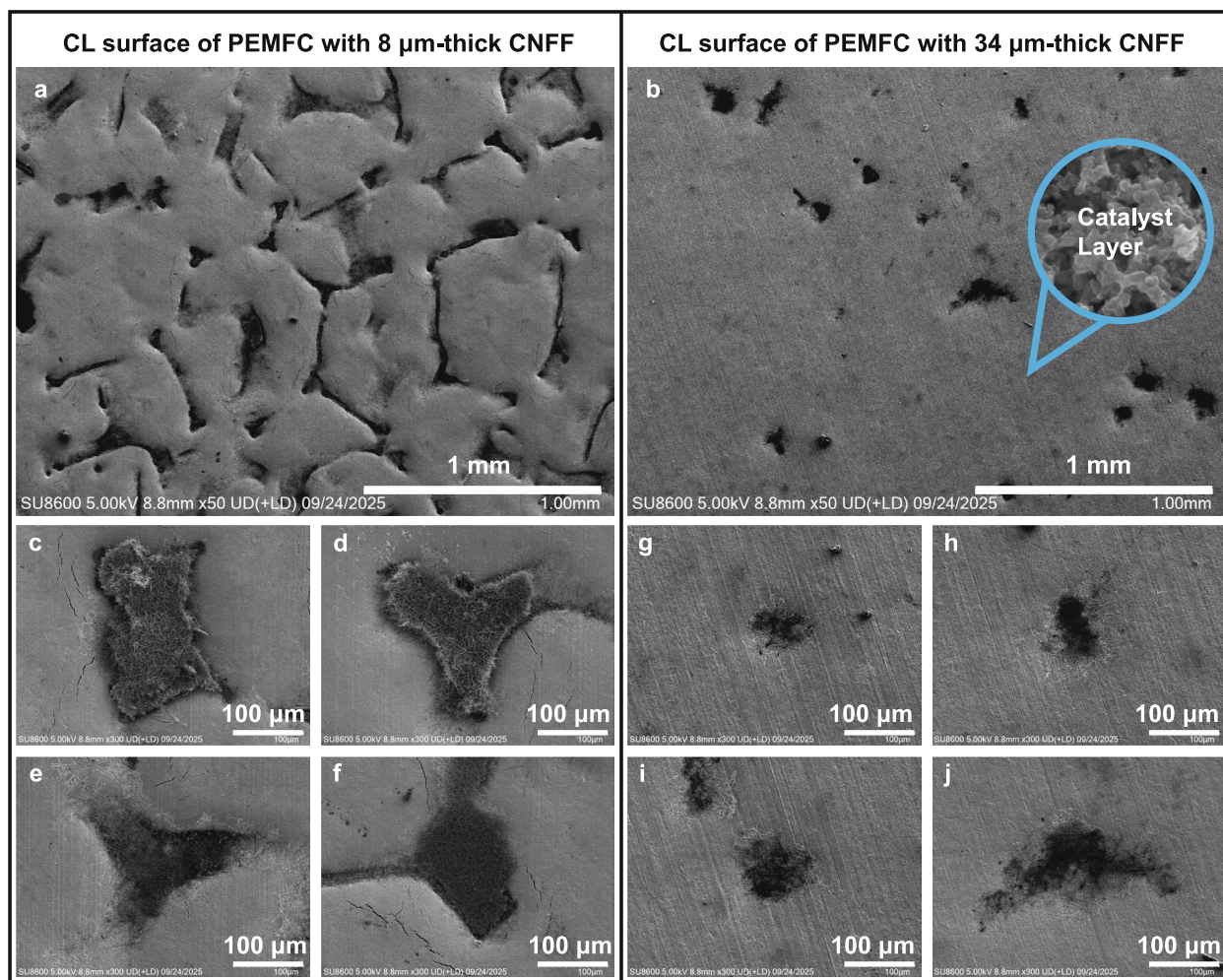


Fig. 3 The SEM images of catalyst layer surfaces using CNFF with different thickness. **a** The CL surface of PEMFC with 8 μm thick CNFF. **b** The CL surface of PEMFC with 34 μm thick CNFF. **c-f** The magnified images of the dents on CL surface and the CNFF residues trapped within using 8 μm thick CNFF. **g-j** The magnified images of the dents on CL surface and the CNFF residues trapped within using 34 μm thick CNFF

monotonically with thickness (8 μm : $2.84 \text{ ms}\cdot\text{m}^{-1}\cdot\text{kPa}^{-1}$; 15 μm : $3.02 \text{ ms}\cdot\text{m}^{-1}\cdot\text{kPa}^{-1}$; 25 μm : $3.14 \text{ ms}\cdot\text{m}^{-1}\cdot\text{kPa}^{-1}$; 34 μm : $3.28 \text{ ms}\cdot\text{m}^{-1}\cdot\text{kPa}^{-1}$), consistent with Fick's law predicting increased resistance in thicker diffusion layers. Conversely, the pressure-independent term decreases within a certain thickness range (8 μm : $0.206 \text{ s}\cdot\text{m}^{-1}$; 15 μm : $0.102 \text{ s}\cdot\text{m}^{-1}$; 25 μm : $0.06 \text{ s}\cdot\text{m}^{-1}$).

Under operating conditions (70 $^{\circ}\text{C}$, 120 kPa), the mean free path of oxygen ($\sim 6.46 \text{ nm}$) is significantly smaller than the CNFF pore size (350 nm–1 μm), rendering Knudsen diffusion in CNFF negligible. Thus, oxygen transport in CNFF is dominated by Fickian diffusion. These results confirm that the increased oxygen transport resistance and concentration overpotential below 25 μm primarily arise from catalyst layer damage induced by the nickel skeleton, which impedes Knudsen diffusion in catalyst layer nanopores and oxygen transport through

polymers, rather than Fickian diffusion blockage due to CNFF collapse. Notably, increasing CNFF thickness within a specific range actively protects the catalyst layer, aligning with ECSA data trends.

3.3 Electrochemical impedance spectroscopy analysis

Figure 5a–c depicts the Nyquist plots and corresponding DRT curves for GDL-less PEMFCs with four CNFF thicknesses (8 μm , 15 μm , 25 μm , 34 μm) at 0.4, 2.0, and 3.0 $\text{A}\cdot\text{cm}^{-2}$. DRT analysis (Fig. 5d–f) reveals overlapping activation and concentration impedances (Peak P3) due to similar relaxation times. At low current density (0.4 $\text{A}\cdot\text{cm}^{-2}$), the 8- μm CNFF exhibits the largest impedance arc, followed by 15 μm , while 25- μm and 34- μm CNFFs show minimal arcs. This trend reflects activation impedance dominance at low currents, where thin CNFF inadequately protects the catalyst layer, leading to reaction

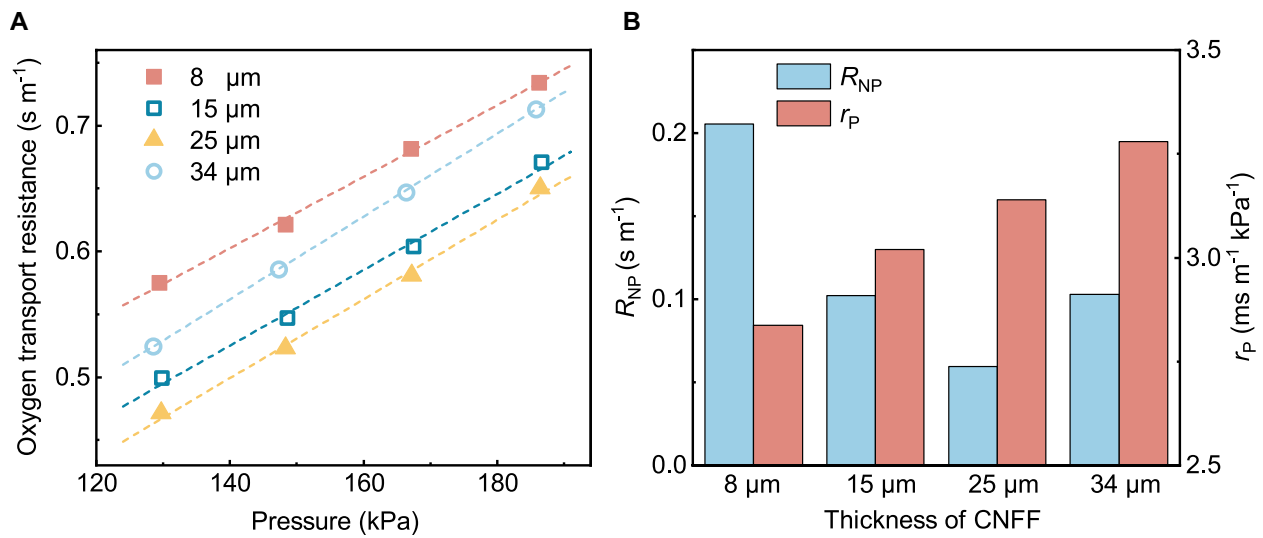


Fig. 4 The thickness of CNFF affects oxygen transport resistance. **a** The trend of the change in oxygen transport resistance with the pressure in the cathode chamber when the GDL-less PEMFC using four different CNFF thicknesses. **b** The influence of CNFF thickness on the pressure-dependent term and pressure-independent term of oxygen transport resistance of the GDL-less PEMFC

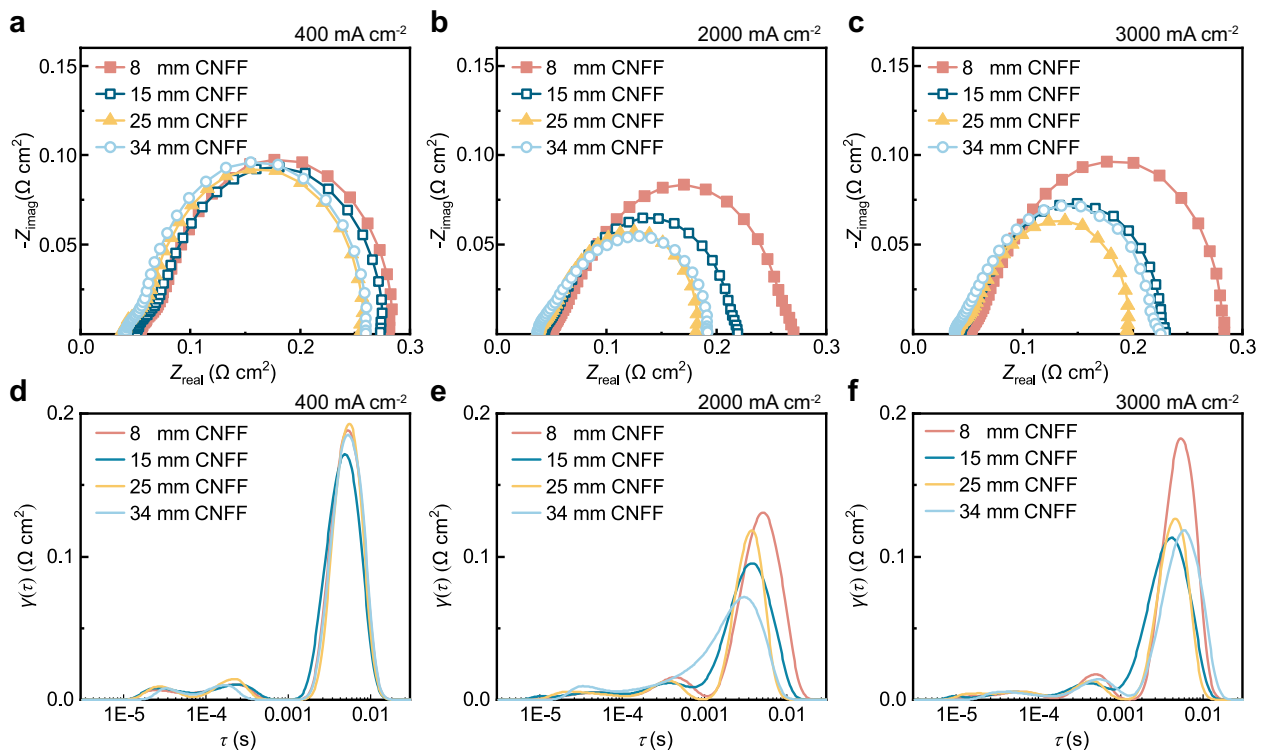


Fig. 5 The EIS of PEMFC with four CNFF thicknesses. Nyquist curves of EIS for ultra-thin integrated fuel cells with four different thicknesses of CNFF under the working current density of **(a)** 0.4 A cm^{-2} , **(b)** 2.0 A cm^{-2} , and **(c)** 3.0 A cm^{-2} respectively. DRT curves of EIS for GDL-less fuel cells with four different thicknesses of CNFF under the working current density of **(d)** 0.4 A cm^{-2} , **(e)** 2.0 A cm^{-2} and **(f)** 3.0 A cm^{-2} respectively

interface deformation under compression and reduced electrochemical active area.

At medium current density ($2.0 \text{ A}\cdot\text{cm}^{-2}$), impedance arcs and relaxation times decrease due to reduced activation impedance with increasing current. Thicker CNFFs exhibit more pronounced reductions, confirming their lower activation impedance. At high current density ($3.0 \text{ A}\cdot\text{cm}^{-2}$), the $34\text{-}\mu\text{m}$ CNFF shows a larger arc than the $25\text{-}\mu\text{m}$ variant due to dominant mass transfer impedance, as excessive thickness prolongs oxygen transport pathways.

The relaxation time distribution contour plot (Fig. 6) demonstrates a non-monotonic trend for Peak P3: relaxation time and peak intensity decrease initially and then increase with rising current density, confirming P3 corresponds to coupled charge transfer and mass transport processes. A critical turning point in relaxation time emerges at high currents, induced by liquid water

accumulation blocking porous structures, reducing the effective diffusion coefficient (D_{eff}) and exacerbating transport limitations [33]. This turning point, indicative of water flooding, shifts to higher current densities with increasing CNFF thickness ($8 \text{ }\mu\text{m}$: $\sim 1.3 \text{ A}\cdot\text{cm}^{-2}$; $15 \text{ }\mu\text{m}$: $\sim 1.6 \text{ A}\cdot\text{cm}^{-2}$; $25 \text{ }\mu\text{m}$: $\sim 2 \text{ A}\cdot\text{cm}^{-2}$; $34 \text{ }\mu\text{m}$: $\sim 2.2 \text{ A}\cdot\text{cm}^{-2}$), confirming that thicker CNFF enhances water management via superior water storage capacity, maintaining oxygen transport pathways even under high water accumulation.

Figure 7 shows the R_{P3} impedance of Peak P3 as a function of current density. From 0.4 to $1 \text{ A}\cdot\text{cm}^{-2}$, R_{P3} decreases with increasing current, with the magnitude of reduction following $8 \text{ }\mu\text{m} < 15 \text{ }\mu\text{m} < 34 \text{ }\mu\text{m} < 25 \text{ }\mu\text{m}$, aligning with activation overpotential behavior. From 1 to $2 \text{ A}\cdot\text{cm}^{-2}$, R_{P3} continues to decrease for all thicknesses except $8 \text{ }\mu\text{m}$, which begins to increase, indicating the onset of mass transport limitations at lower currents.

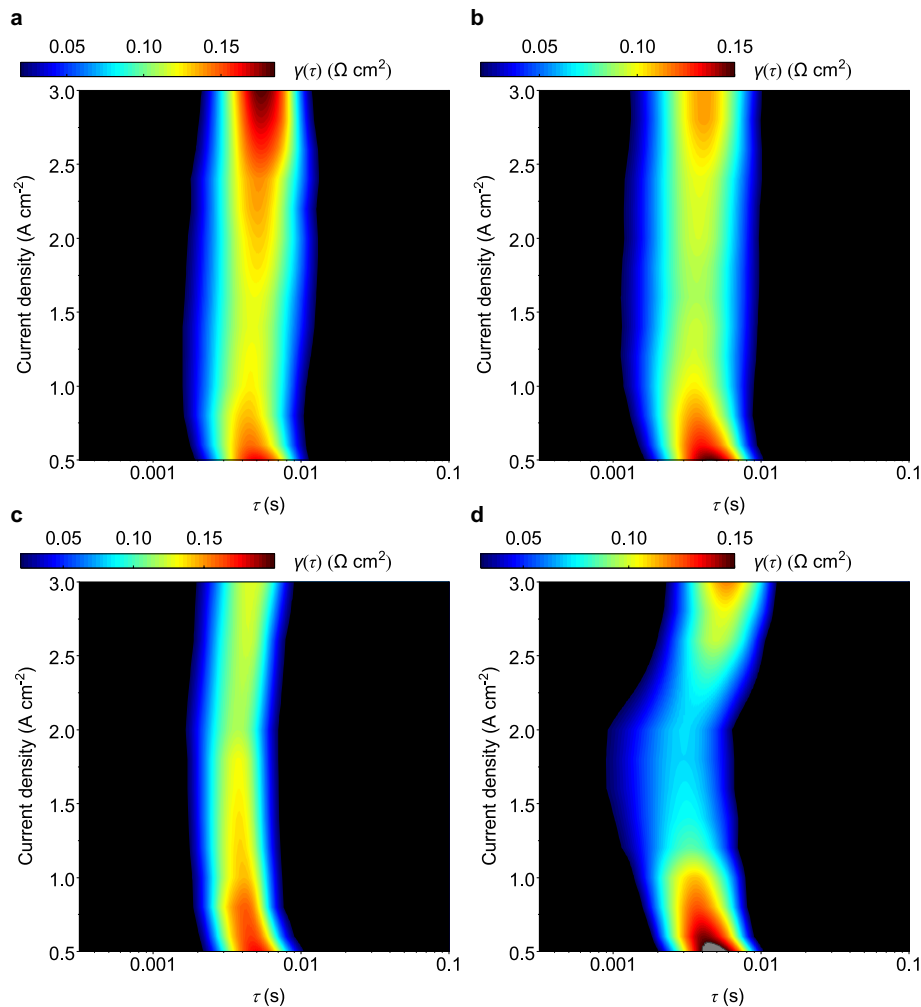


Fig. 6 Variation of relaxation time with current density. Relaxation time distribution contour plots of GDL-less PEMFCs with four CNFF thicknesses, showing variations with current density: (a) $8 \text{ }\mu\text{m}$; (b) $15 \text{ }\mu\text{m}$; (c) $25 \text{ }\mu\text{m}$; (d) $34 \text{ }\mu\text{m}$

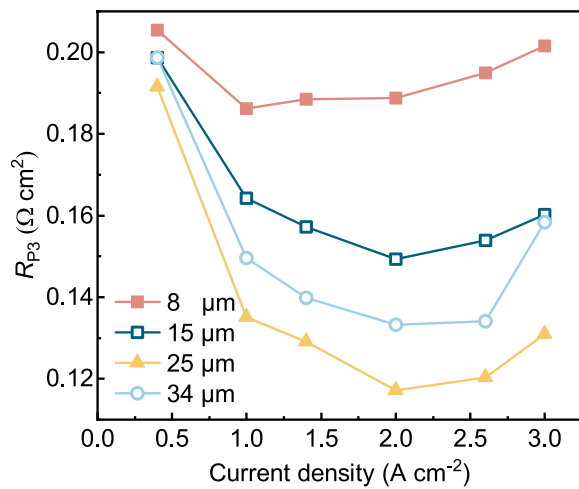


Fig. 7 The change of the impedance value corresponding to the P3 peak in the GDL-less PEMFCs with four different thicknesses of CNFF respectively as the current density varies

Between 2 and 2.6 $\text{A}\cdot\text{cm}^{-2}$, R_{p3} increases for all thicknesses due to high-current mass transport losses, with the rate of increase diminishing as thickness increases. For example, the 34- μm CNFF exhibits better water management than the 15- μm variant despite higher oxygen transport resistance, attributed to its superior water storage capacity. From 2.6 to 3 $\text{A}\cdot\text{cm}^{-2}$, R_{p3} increases more rapidly with thickness as liquid water fills pores in thicker CNFFs, re-establishing concentration impedance dominance.

3.4 Dynamic pressure drops

Figure 8a illustrates the relationship between the inlet pressures of the anode and cathode in the ultra-thin GDL-less PEMFC and the voltage. For the anode, a significant correlation exists: the inlet pressure decreases as the voltage drops and increases as the voltage rises. Notably, the inflection points of the anode inlet pressure and voltage changes coincide precisely. This is because as the voltage decreases, the current density increases, accelerating the consumption of reaction gas within the PEMFC, thus causing the inlet pressure to decline.

In contrast, for the cathode, as the voltage decreases and the current density increases, more liquid water is generated. This liquid water infiltrates the pores of the metal foam, increasing the pressure difference across the flow field [34]. Significantly, the inflection point of the cathode inlet pressure change lags behind that of the voltage change. The time interval between their peaks defines this lag time, which arises due to the time required for the reaction-generated liquid water to enter the metal foam and impact the inlet pressure drop. The lag time primarily depends on two factors: the liquid water generation rate (i.e., current density) and the liquid water storage capacity of the CNFF layer.

To investigate the effect of CNFF thickness on the transient change characteristics of cathode pressure with voltage, the cathode pressure-voltage relationships of GDL-less PEMFCs with different CNFF thicknesses were examined, as depicted in Fig. 8b. During the linear voltage scan from the lowest to the highest point,

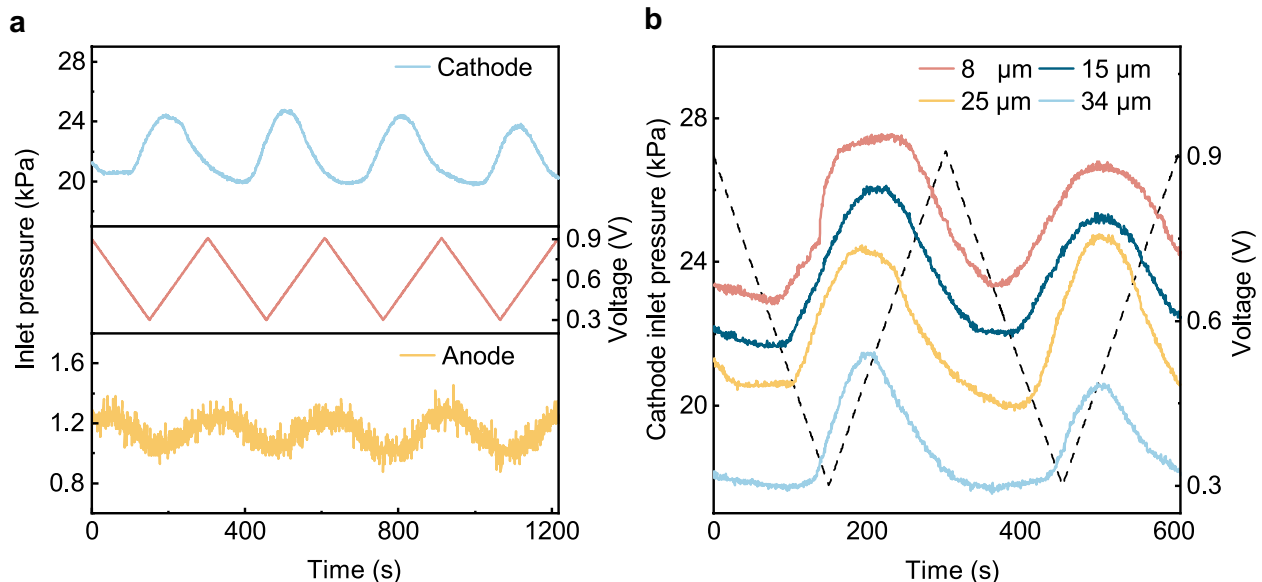


Fig. 8 The thickness of CNFF affects water management capability. The transient characteristics of the inlet pressure of the ultra-thin integrated fuel cell varying with the voltage, (a) Comparison of the inlet pressures of the cathode and the anode; (b) Comparison of the cathode inlet pressures corresponding to four CNFF thicknesses

the upward trend of the cathode inlet pressure gradually reverses and decreases. Thicker CNFF layers exhibit earlier inflection points, indicating superior water storage capabilities. Conversely, during the scan from the highest to the lowest voltage, the inlet pressure reaches a minimum. Thicker CNFF layers delay the inflection point and prolong the low-pressure state. Owing to their stronger water storage capacity, thicker CNFF layers can retain more liquid water before it is discharged into the nickel foam, thereby effectively reducing the pressure drop increase caused by liquid water occupying the pores of the nickel foam. Additionally, studies have demonstrated that better water management capability corresponds to a more uniform reactant distribution and less degradation of the catalyst layer and polymer, thereby effectively enhancing PEMFC lifespan and long-term stability [35, 36].

4 Conclusion

This study investigates the effect of CNFF thickness on ultrathin integrated fuel cells. CNFF samples (8 μm , 15 μm , 25 μm , 34 μm) are prepared by adjusting spinning time (other electrospinning parameters like humidity and voltage fixed), then integrated into the fuel cells. Polarization curves, oxygen transport resistance measurements, cyclic voltammetry, high-frequency impedance analysis, electrochemical impedance spectroscopy, and inlet–outlet pressure drop tests are used to analyze performance variations and mechanisms.

Polarization curves show notable performance differences among cells with varied CNFF thicknesses: the 25- μm CNFF cell has the highest peak power density (1.31 $\text{W}\cdot\text{cm}^{-2}$), the 8- μm one the lowest. Peak power density rises with thickness within a range but drops beyond 25 μm , indicating an optimal thickness. Analysis of three overpotentials reveals ohmic and activation overpotentials decrease monotonically with thickness, while concentration overpotential first falls then rises. The performance degradation at 34 μm is mainly due to higher concentration overpotential. ECSA measurements show thinner CNFF has smaller ECSA; with identical catalyst layers, thinner CNFF fails to protect the catalyst layer from nickel foam skeleton damage, explaining ohmic/activation overpotential trends.

Oxygen transport resistance changes non-monotonically with CNFF thickness: the pressure-dependent component increases (longer diffusion paths), while the pressure-independent component first decreases then increases (consistent with ECSA results, as thinner CNFF causes worse catalyst layer damage). Since the pressure-independent term is primarily governed by Knudsen diffusion in the nanopores of the catalyst layer and oxygen

transfer in the polymer, thinner CNFF leads to higher resistance, clarifying concentration overpotential trends.

EIS shows τ_{p3} and R_{p3} , associated with cathode charge transfer and oxygen mass transport, first decrease then increase with current density; the turning point shifts to higher current densities with thicker CNFF. Transient pressure drop tests indicate thicker CNFF has greater water storage, enhancing resistance to flooding at high current densities and delaying concentration impedance degradation. These results explain concentration loss trends.

In summary, this study elucidates the impact of CNFF thickness on ultrathin GDL-less PEMFCs, clarifies the mechanisms of different overpotentials, and proposes thickness optimization strategies to further improve performance.

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

Chasen Tongsh, Siyuan Wu, Kui Jiao conceived of the idea, designed, and performed the experiments. Chasen Tongsh, Siyuan Wu carried out the sample fabrication. Chasen Tongsh, Siyuan Wu, Kui Jiao, Daokuan Jiao analyzed the experimental data. Chasen Tongsh, Siyuan Wu wrote the manuscript. Kui Jiao provided supervision and feedback, and edited the manuscript. All the authors analyzed and verified the results, and commented on the manuscript.

Data Availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests

The authors declare no competing interests.

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